



PROJECT AND TEAM INFORMATION

Project Title

OS User Activity Monitoring & Insider Threat Detection System

Team information	Details
Team name and number	KernelGuard OSDBMS-V-2026-T298
Mentor	(Prof.)Dr.Mohammad Wazid
Team member 1 (Team Lead)	Arshpreet Singh 24021835 24021835@geu.ac.in 
Team member 2	Mohd Ahmed Khan 24021365 24021365@geu.ac.in 
Team member 3	Ankit 240212134 240212134@geu.ac.in 
Team member 4	Mohd Shoaib 240211124 240211124@geu.ac.in 

PHASE-II PROJECT PROGRESS DESCRIPTION (35 pts)

Project Abstract (2 pts)

KernelGuard Phase II demonstrates a traceable path from Linux Audit output to an administrator alert. A collector groups related audit records, extracts selected identity and file fields, and stores them in a dedicated MySQL database. Two rules identify repeated authentication failures and successful protected-file access outside configured hours. The dashboard shows each alert with its supporting events and policy values. The nine-event fixture is clearly labelled synthetic, while live records carry a separate origin.

Updated Project Approach and Architecture (2 pts)

A Linux audit rule watches a protected directory. The parser groups lines by audit event identity and normalizes timestamps, outcome, UID fields and path metadata. A unique source identifier prevents duplicate imports. The collector obtains a MySQL writer lock and stores events, alerts and its checkpoint in one transaction. SQLAlchemy defines eight tables with keys, foreign keys and indexes. Flask serves a localhost dashboard through a SELECT-only database account. A separate fixture imports controlled events for repeatable classroom evaluation. This phase deliberately limits detection to two rules so the evidence path can be inspected end to end.

Tasks Completed (7 pts)

Task Completed	Team Member
Audit rule, parser and checkpoint collector	Arshpreet Singh
Eight-table MySQL schema and SQL queries	Mohd Ahmed Khan
Two policies and fixture checks	Ankit
Read-only dashboard and alert evidence pages	Mohd Shoaib

Challenges/Roadblocks (7 pts)

Linux Audit emits several lines for one action, so the parser must form a complete event before applying a rule. Collector restarts can replay records; unique source IDs and a checkpoint in the write transaction address that case. The fixture makes demonstrations reproducible but cannot establish live collection. A controlled protected-file read and MySQL 8.4 integration check were therefore recorded separately on the Kali lab host.

Tasks Pending (7 pts)

Task Pending	Team Member
User, session, process and USB inventory	Phase III
Three additional policy families and review history	Phase III
Physical USB acceptance check	Phase III

Project Outcome/Deliverables (2 pts)

The Phase II deliverable is a working two-rule Linux Audit and MySQL prototype. It includes the collection rule, parser, eight-table relational model, fixture, SQL analysis queries, read-only dashboard and tests. The fixture produces nine events, two alerts and six evidence links. A Kali check dated 2 October 2026 also recorded a live protected-file read and a passing Oracle MySQL 8.4 integration run.

Progress Overview (2 pts)

The audit-to-alert path and local dashboard are implemented. The saved Kali check covers a controlled live file read and the MySQL-backed workflow. Broader host inventory, privileged-command and bulk-file rules, USB enrollment, and administrator decisions belong to the Phase III scope. Phase II results should be presented as a focused prototype; fixture counts are synthetic and do not measure field detection accuracy.

Codebase Information (2 pts)

Local source: C:\KernalGuard-Phase2-PBL. Branch: main. This standalone copy has no commit history or configured public remote. Main files: app.py, kernelguard/parser.py, kernelguard/core.py, kernelguard/locking.py, linux/kernelguard.rules, sql/schema.mysql.sql and tests/test_app.py. Technical basis: Red Hat audit-log documentation, SQLAlchemy transaction documentation and the MySQL 8.4 foreign-key manual.

Testing and Validation Status (2 pts)

Test Type	Status	Notes
Local automated tests	3 passed, 1 skipped	MySQL test is opt-in
MySQL 8.4 integration	4 passed	Recorded Kali run, 2 Oct 2026
Controlled protected read	Pass	Live audit row recorded on Kali

Deliverables Progress (2 pts)

Completed: collector, parser, two policies, relational schema, SQL queries, fixture, dashboard and setup guide. Demonstrated separately: live Kali read and MySQL 8.4 integration. The standalone Phase II copy documents this milestone; it is not a historical Git snapshot of the original submission.